

# Homework 3

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## Problem 1.

*Proof.* Let  $\gamma_R$ , be a semicircle contour of radius  $R$ .

$$\lim_{R \rightarrow \infty} \int_{\gamma_R} \left( \frac{e^{iz}}{z^2 + a^2} \right) dz = \lim_{R \rightarrow \infty} \int_0^{2\pi} \left( \frac{Rie^{i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right) d\theta + \lim_{R \rightarrow \infty} \int_{-R}^R \frac{e^{ix}}{x^2 + a^2} dx. \quad (1)$$

Focusing on the arc component of the contour, by the integral equality

$$\begin{aligned} \left| \int_0^{2\pi} \left( \frac{Rie^{i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right) d\theta \right| &\leq \int_0^{2\pi} \left| \frac{Rie^{i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right| d\theta \\ &= \int_0^{2\pi} \frac{|R|}{|R^2 e^{2i\theta} + a^2|} |e^{iRe^{i\theta}}| d\theta \\ &\leq \int_0^{2\pi} \frac{|R|}{|R^2 e^{2i\theta}|} |e^{iRe^{i\theta}}| d\theta \\ &= \int_0^{2\pi} \frac{1}{|R|} |e^{iRe^{i\theta}}| d\theta = \int_0^{2\pi} \frac{1}{|R|} |e^{-\sin(\theta)R} e^{iR\cos(\theta)}| d\theta \\ &= \int_0^{2\pi} \left| \frac{e^{-R\sin(\theta)}}{R} \right| d\theta \end{aligned}$$

Thus as  $R \rightarrow \infty$ ,  $e^{-R\sin(\theta)} \rightarrow 0$  and  $\frac{1}{|R|} \rightarrow 0$ . Thus  $\int_0^{2\pi} \left| \frac{e^{-R\sin(\theta)}}{R} \right| d\theta \rightarrow 0$ . Thus,

$$\lim_{R \rightarrow \infty} \int_0^{2\pi} \left( \frac{Rie^{i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right) d\theta = 0 \quad (2)$$

Thus,

$$\lim_{R \rightarrow \infty} \int_{\gamma_R} \left( \frac{e^{iz}}{z^2 + a^2} \right) dz = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{e^{ix}}{x^2 + a^2} dx = \int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + a^2} dx.$$

$z^2 + a^2 = (z - ia)(z + ia)$ , thus  $\frac{e^{iz}}{z^2 + a^2}$  only has poles at  $z = \pm ia$ . Only  $ia$  is inside the semicircle bounded by our contour.

$$(z - ia) \frac{e^{iz}}{z^2 + a^2} = \frac{e^{iz}}{z + ia}$$

$\frac{e^{iz}}{z+ia}$  is holomorphic on the domain bounded by the contour since  $e^{iz}$  is holomorphic and  $z+ia$  is holomorphic and nonzero. Thus  $z=ia$  is a simple pole.

Thus by residue theorem and since  $z=ia$  is the only pole in the domain bounded by the contour,

$$\begin{aligned}\lim_{R \rightarrow \infty} \oint_{\gamma_R} \left( \frac{e^{iz}}{z^2 + a^2} \right) dz &= 2\pi i \lim_{z \rightarrow ia} (z - ia) \frac{e^{iz}}{z^2 + a^2} \\ &= 2\pi i \lim_{z \rightarrow ia} \frac{e^{iz}}{z + ia} \\ &= \frac{2\pi i e^{-a}}{2ia} \\ &= \frac{\pi e^{-a}}{a}\end{aligned}$$

Thus by 2,

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2 + a^2} dx &= \frac{i\pi a e^{-a}}{a} \\ &= \frac{\pi e^{-a}}{a}.\end{aligned}$$

Since,

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{x \sin(x)}{x^2 + a^2} dx &= \Im \left( \int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2 + a^2} dx \right) \\ &= \Im(i\pi e^{-a}) = \frac{\pi e^{-a}}{a}\end{aligned}$$

Thus,

$$\int_{-\infty}^{\infty} \frac{x \sin(x)}{x^2 + a^2} dx = \pi e^{-a}$$

□

## Problem 2.

*Proof.* Let  $\gamma_R$ , be a semicircle contour of radius  $R$ .

$$\lim_{R \rightarrow \infty} \int_{\gamma_R} \left( \frac{z e^{iz}}{z^2 + a^2} \right) dz = \lim_{R \rightarrow \infty} \int_0^{2\pi} \left( \frac{R^2 i e^{2i\theta} e^{iR e^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right) d\theta + \lim_{R \rightarrow \infty} \int_{-R}^R \frac{x e^{ix}}{x^2 + a^2} dx. \quad (3)$$

Focusing on the arc component of the contour, by the integral equality

$$\begin{aligned}
\left| \int_0^{2\pi} \left( \frac{R^2 i e^{2i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right) d\theta \right| &\leq \int_0^{2\pi} \left| \frac{R^2 i e^{2i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right| d\theta \\
&= \int_0^{2\pi} \frac{|R^2|}{|R^2 e^{2i\theta} + a^2|} |e^{iRe^{i\theta}}| d\theta \\
&\leq \int_0^{2\pi} \frac{|R^2|}{|R^2 e^{2i\theta}|} |e^{iRe^{i\theta}}| d\theta \\
&= \int_0^{2\pi} |e^{iRe^{i\theta}}| d\theta = \int_0^{2\pi} |e^{-\sin(\theta)R} e^{iR \cos(\theta)}| d\theta \\
&= \int_0^{2\pi} |e^{-R \sin(\theta)}| d\theta
\end{aligned}$$

Thus as  $R \rightarrow \infty$ ,  $e^{-R \sin(\theta)} \rightarrow 0$ . Thus  $\int_0^{2\pi} |e^{-R \sin(\theta)}| d\theta \rightarrow 0$ , meaning:

$$\lim_{R \rightarrow \infty} \int_0^{2\pi} \left( \frac{R^2 i e^{2i\theta} e^{iRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} \right) d\theta = 0 \quad (4)$$

Thus,

$$\lim_{R \rightarrow \infty} \int_{\gamma_R} \left( \frac{ze^{iz}}{z^2 + a^2} \right) dz = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{xe^{ix}}{x^2 + a^2} dx = \int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2 + a^2} dx. \quad (5)$$

$z^2 + a^2 = (z - ia)(z + ia)$ , thus  $\frac{ze^{iz}}{z^2 + a^2}$  only has poles at  $z = \pm ia$ . Only  $ia$  is inside the semicircle bounded by our contour.

$$(z - ia) \frac{ze^{iz}}{z^2 + a^2} = \frac{ze^{iz}}{z + ia}$$

$\frac{ze^{iz}}{z + ia}$  is holomorphic on the domain bounded by the contour since  $ze^{iz}$  is holomorphic and  $z + ia$  is holomorphic and nonzero. Thus  $z = ia$  is a simple pole.

Thus by residue theorem and since  $z = ia$  is the only pole in the domain bounded by the contour,

$$\begin{aligned}
\lim_{R \rightarrow \infty} \oint_{\gamma_R} \left( \frac{ze^{iz}}{z^2 + a^2} \right) dz &= 2\pi i \lim_{z \rightarrow ia} (z - ia) \frac{ze^{iz}}{z^2 + a^2} \\
&= 2\pi i \lim_{z \rightarrow ia} \frac{ze^{iz}}{z + ia} \\
&= \frac{-2\pi a e^{-a}}{2ia} \\
&= \frac{i\pi a e^{-a}}{a}
\end{aligned}$$

Thus by 5,

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2 + a^2} dx &= \frac{i\pi a e^{-a}}{a} \\ &= i\pi e^{-a}.\end{aligned}$$

Since,

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{x \sin(x)}{x^2 + a^2} dx &= \Im \left( \int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2 + a^2} dx \right) \\ &= \Im(i\pi e^{-a}) = \pi e^{-a}\end{aligned}$$

Thus,

$$\int_{-\infty}^{\infty} \frac{x \sin(x)}{x^2 + a^2} dx = \pi e^{-a} \quad (6)$$

□

### Problem 8.

*Proof.* Let  $\gamma$  be contour along the unit circle. Let  $\alpha = \frac{b}{a}$ . Let  $\mathbb{D} := \{z \in \mathbb{C} \mid |z| < 1\}$ .

Let  $\lambda_+ := \frac{-1 + \sqrt{1 - \alpha^2}}{\alpha}$  and let  $\lambda_- = \frac{-1 - \sqrt{1 - \alpha^2}}{\alpha}$ .

Since  $a > |b| \implies |\alpha| < 1 \implies \left| \frac{1}{\alpha} \right| > 1$ .

$$\begin{aligned}\lambda_- \lambda_+ &= \left( \frac{-1 + \sqrt{1 - \alpha^2}}{\alpha} \right) \left( \frac{-1 - \sqrt{1 - \alpha^2}}{\alpha} \right) \\ &= \frac{1 - (1 - \alpha^2)}{\alpha} = \alpha\end{aligned}$$

1. **If  $b < 0$ :** Thus  $\alpha < 0$  and  $\frac{1}{\alpha} < 0$ . Thus  $\frac{1}{\alpha} < -1 < 0 \implies -\frac{1}{\alpha} > 1$ . Since  $\sqrt{1 - \alpha^2} > 0 \implies -\frac{\sqrt{1 - \alpha^2}}{\alpha} > 0$ . Thus, since  $-\frac{\sqrt{1 - \alpha^2}}{\alpha} < 0$ ,  $\lambda_- = \frac{-1}{\alpha} - \frac{\sqrt{1 - \alpha^2}}{\alpha} > 1$ . Thus  $\lambda_- \notin \mathbb{D}$ .

2. **If  $b > 0$ :** Thus  $\alpha > 0$  and  $\frac{1}{\alpha} > 0$ . Thus  $\frac{1}{\alpha} > 1 \implies -\frac{1}{\alpha} < 0$ . Since  $\sqrt{1 - \alpha^2} > 0 \implies -\frac{\sqrt{1 - \alpha^2}}{\alpha} < 0$ . Thus, since  $-\frac{\sqrt{1 - \alpha^2}}{\alpha} < 0$ ,  $\lambda_- = -\frac{1}{\alpha} - \frac{\sqrt{1 - \alpha^2}}{\alpha} < -\frac{1}{\alpha} < -1$ . Thus  $\lambda_- \notin \mathbb{D}$ .

But since  $\lambda_- \lambda_+ = \alpha$ , and  $|\alpha| < 1$  and  $|\lambda_-| > 1$ ,  $|\lambda_+| < \alpha < 1$ . Thus  $\lambda_+ \in \mathbb{D}$

Since the zeros of  $\alpha z^2 + 2z + \alpha$  are  $\lambda_+$  and  $\lambda_-$ ,  $\frac{1}{\alpha z^2 + 2z + \alpha}$  has no other poles other than  $\lambda_+$  and  $\lambda_-$ .

$$\begin{aligned}(z - \lambda_+) \frac{1}{\alpha z^2 + 2z + \alpha} &= (z - \lambda_+) \frac{1}{\alpha(z - \lambda_+)(z - \lambda_-)} \\ &= \frac{1}{z - \lambda_-}\end{aligned}$$

and  $\frac{1}{z - \lambda_-}$  is holomorphic on the unit disk. Thus  $z = \lambda_+$  is simple pole of  $\frac{1}{\alpha z^2 + 2z + \alpha}$ .

Thus by the residue theorem,

$$\begin{aligned}
\oint_{\gamma} \frac{1}{\alpha z^2 + 2z + \alpha} dz &= 2\pi i \operatorname{res}_{\lambda_+} \left( \frac{1}{\alpha z^2 + 2z + \alpha} \right) \\
&= 2\pi i \lim_{z \rightarrow \lambda_+} \frac{(z - \lambda_+)}{\alpha(z - \lambda_+)(z - \lambda_-)} \\
&= \frac{2\pi i}{\alpha(\lambda_+ - \lambda_-)} \\
&= \frac{2\pi i}{2\sqrt{1 - \alpha^2}} \\
&= \frac{\pi i}{\sqrt{1 - \alpha^2}}
\end{aligned}$$

$$\begin{aligned}
\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} &= \frac{1}{a} \int_0^{2\pi} \frac{d\theta}{1 + \alpha \cos \theta} \\
&= \frac{1}{a} \int_0^{2\pi} \frac{d\theta}{1 + \frac{\alpha e^{i\theta} + \alpha e^{-i\theta}}{2}} \\
&= \frac{2}{a} \int_0^{2\pi} \frac{d\theta}{2 + \alpha e^{i\theta} + \alpha e^{-i\theta}} \\
&= \frac{2}{a} \int_0^{2\pi} \frac{d\theta}{2 + \alpha e^{i\theta} + \frac{\alpha}{e^{i\theta}}} \\
&= \frac{2}{a} \oint_{\gamma} \frac{1}{zi(2 + \alpha z + \frac{\alpha}{z})} dz \\
&= \frac{2}{ai} \oint_{\gamma} \frac{1}{\alpha z^2 + 2z + \alpha} dz \\
&= \frac{2}{ai} \left( \frac{\pi i}{\sqrt{1 - \alpha^2}} \right) \\
&= \frac{2\pi}{a\sqrt{1 - \frac{b^2}{a^2}}} \\
&= \frac{2\pi b}{\sqrt{a^2 - b^2}}
\end{aligned}$$

□

### Problem 9.

*Proof.* Let  $\gamma_R$  be the rectangle contour with corners at 0, 1,  $1 + Ri$ , and  $Ri$ .

Slit the plane on  $(\infty, 0]$  and define the domain of the complex logarithm as  $\mathbb{C} - (-\infty, 0]$ .

Let  $z = a + bi$ . By the sum and difference identities of sine and cosine and definition of hyperbolic trigonometric functions,

$$\begin{aligned}\sin(\pi z) &= \sin(\pi a + \pi bi) \\ &= \sin(\pi a) \cos(\pi bi) + \cos(\pi a) \sin(\pi bi) \\ &= \sin(\pi a) \cosh(\pi b) + i \cos(\pi a) \sinh(\pi b)\end{aligned}$$

By Cauchy's Theorem,

$$\lim_{R \rightarrow \infty} \oint_{\gamma} \log(\sin(\pi z)) dz = 0 \quad (7)$$

However, the contour integral also equals

$$\begin{aligned}\lim_{R \rightarrow \infty} \oint_{\gamma} \log(\sin(\pi z)) dz &= \int_0^1 \log(\sin(\pi x)) dx \\ &\quad + \lim_{R \rightarrow \infty} i \int_0^R \log(\sin(\pi(1 + iy))) dy \\ &\quad + \lim_{R \rightarrow \infty} \int_1^0 \log(\sin(\pi(x + iR))) dx \\ &\quad + \lim_{R \rightarrow \infty} i \int_R^0 \log(\sin(i\pi y)) dy\end{aligned}$$

Doing the vertical components:

$$\begin{aligned}\lim_{R \rightarrow \infty} i \left[ \int_0^R \log(\sin(\pi(1 + iy))) dy + \int_R^0 \log(\sin(i\pi y)) dy \right] &= \lim_{R \rightarrow \infty} \left[ i \int_0^R \log(\sin(\pi(1 + iy))) dy + i \int_R^0 \log(\sin(i\pi y)) dy \right] \\ &= \lim_{R \rightarrow \infty} \left[ i \int_0^R \log(-i \sinh(y\pi)) dy + i \int_R^0 \log(i \sinh(\pi y)) dy \right] \\ &= \lim_{R \rightarrow \infty} i \int_0^R \left( \log(\sinh(y\pi)) - \frac{\pi}{2} i \right) dy \\ &\quad - \lim_{R \rightarrow \infty} i \int_0^R \left( \log(\sinh(\pi y)) + \frac{\pi}{2} i \right) dy \\ &= \lim_{R \rightarrow \infty} \int_0^R \left( \frac{\pi}{2} + \frac{\pi}{2} \right) dy \\ &= \lim_{R \rightarrow \infty} \pi R\end{aligned}$$

Doing the top horizontal component:

$$\begin{aligned}
\lim_{R \rightarrow \infty} \int_1^0 \log(\sin(\pi(x + iR))) dx &= - \lim_{R \rightarrow \infty} \int_0^1 \log(\sin(\pi x) \cosh(R\pi) + i \cos(\pi x) \sinh(R\pi)) dx \\
&= - \lim_{R \rightarrow \infty} \int_0^1 \log \left[ \sin(\pi x) \left( \frac{e^{R\pi} + e^{-R\pi}}{2} \right) + i \cos(\pi x) \left( \frac{e^{R\pi} - e^{-R\pi}}{2} \right) \right] dx \\
(\text{For large } R) &= - \lim_{R \rightarrow \infty} \int_0^1 \log \left[ \sin(\pi x) \left( \frac{e^{R\pi}}{2} \right) + i \cos(\pi x) \left( \frac{e^{R\pi}}{2} \right) \right] dx \\
&= - \lim_{R \rightarrow \infty} \int_0^1 \log \left[ \frac{e^{R\pi}}{2} (\sin(\pi x) + i \cos(\pi x)) \right] dx \\
&= - \lim_{R \rightarrow \infty} \int_0^1 \log \left[ \frac{e^{R\pi}}{2} e^{i\pi(\frac{1}{2} - x)} \right] dx \\
&= - \lim_{R \rightarrow \infty} \int_0^1 \left( \log \frac{e^{R\pi}}{2} + \frac{\pi}{2} - \pi x \right) dx \\
&= - \lim_{R \rightarrow \infty} \int_0^1 \left( R\pi - \log(2) + \frac{\pi}{2} - \pi x \right) dx \\
&= - \lim_{R \rightarrow \infty} \left( R\pi - \log(2) + \frac{\pi}{2} - \frac{\pi}{2} \right) \\
&= - \lim_{R \rightarrow \infty} (R\pi - \log(2))
\end{aligned}$$

Thus,

$$\begin{aligned}
0 &= \lim_{R \rightarrow \infty} \oint_{\gamma} \log(\sin(\pi z)) dz \\
&= \int_0^1 \log(\sin(\pi x)) dx \\
&\quad + \lim_{R \rightarrow \infty} \left[ \int_0^R \log(\sin(\pi(1 + iy))) dy + \int_R^0 \log(\sin(i\pi y)) dy \right] \\
&\quad + \lim_{R \rightarrow \infty} \int_1^0 \log(\sin(\pi(x + iR))) dx \\
&= \lim_{R \rightarrow \infty} \int_0^1 \log(\sin(\pi x)) dx + \lim_{R \rightarrow \infty} (R\pi - R\pi + \log(2)) \\
0 &= \lim_{R \rightarrow \infty} \int_0^1 \log(\sin(\pi x)) dx + \log(2)
\end{aligned}$$

Thus,

$$\int_0^1 \log(\sin(\pi x)) dx = -\log(2)$$

□

**Problem 10.**

*Proof.* Let  $\gamma_{R,\epsilon}$  be the semicircle cutout contour with a radius of  $R$  and a inner radius of  $\epsilon$ . Slit the plane on  $\{i(-\infty, 0]\}$  and define  $\mathbb{C} - \{i(\infty, 0]\}$  as the domain of the complex logarithm.

Let

$$f(x) = \frac{\log(x)}{x^2 + a^2} = \frac{\log(x)}{(x - ia)(x + ia)} = \frac{1}{x - ia} \left( \frac{\log(x)}{x + ia} \right)$$

By Cauchy's formula,

$$\lim_{R \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \oint_{\gamma_{R,\epsilon}} f(x) dx = \lim_{R \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \oint_{\gamma_{R,\epsilon}} \frac{1}{x - ia} \left( \frac{\log(x)}{x + ia} \right) dx = \lim_{R \rightarrow \infty} \lim_{\epsilon \rightarrow 0} 2\pi i \left( \frac{\log(ia)}{ia + ia} \right) = \frac{\pi}{a} \left[ \log(a) + \frac{\pi}{2} i \right]$$

However the contour integral is also equal to,

$$\oint_{\gamma_{R,\epsilon}} f(x) dx = \int_{\epsilon}^R f(x) dx + \int_0^{\pi} f(Re^{i\theta}) iRe^{i\theta} d\theta + \int_{-R}^{-\epsilon} f(x) dx + \int_{\pi}^0 f(\epsilon e^{i\theta}) i\epsilon e^{i\theta} d\theta \quad (8)$$

By taking the portions on the real axis,

$$\lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R f(x) dx + \int_{-R}^{-\epsilon} f(x) dx \right] = \lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R \frac{\log(x)}{x^2 + a^2} dx + \int_{-R}^{-\epsilon} \frac{\log(-x) + i\pi}{x^2 + a^2} dx \right] \quad (9)$$

$$= \lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R \frac{\log(x)}{x^2 + a^2} dx - \int_R^{\epsilon} \frac{\log(u) + i\pi}{u^2 + a^2} du \right] \quad (10)$$

$$= \lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R f(x) dx + \int_{\epsilon}^R f(x) + i\pi \int_{\epsilon}^R \frac{1}{x^2 + a^2} dx \right] \quad (11)$$

$$= 2 \lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R f(x) dx + \frac{i\pi}{a} \left[ \arctan\left(\frac{x}{a}\right) \right]_{\epsilon}^R \right] \quad (12)$$

$$= 2 \lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R f(x) dx + \frac{i\pi}{a} (\arctan(R) + \arctan(\epsilon)) \right] \quad (13)$$

$$= 2 \lim_{R \rightarrow \infty, \epsilon \rightarrow 0} \left[ \int_{\epsilon}^R f(x) dx \right] + \frac{i\pi^2}{2a}. \quad (14)$$

Line 14 is valid because as  $R \rightarrow \infty$ ,  $\arctan(R) \rightarrow \frac{\pi}{2}$  and as  $\epsilon \rightarrow 0$ ,  $\arctan(\epsilon) \rightarrow 0$ .

By taking the curve on the a semicircle with radius  $R$ ,

$$\begin{aligned} \lim_{R \rightarrow \infty} \int_0^{\pi} f(Re^{i\theta}) iRe^{i\theta} d\theta &= i \lim_{R \rightarrow \infty} \int_0^{\pi} \frac{Re^{i\theta} \log(Re^{i\theta})}{R^2 e^{2i\theta} + a^2} d\theta \\ &= i \lim_{R \rightarrow \infty} \int_0^{\pi} \frac{(e^{i\theta} \log(R) + ie^{i\theta} \theta)}{Re^{2i\theta} + \frac{a^2}{R}} d\theta \end{aligned}$$

As  $R \rightarrow \infty$ ,  $R$  grows faster than  $\log(R)$  thus  $R$  dominates meaning the denominator dominates, and  $\frac{(e^{i\theta} \log(R) + ie^{i\theta} \theta)}{Re^{2i\theta} + \frac{a^2}{R}} \rightarrow 0$

Thus:

$$\lim_{R \rightarrow \infty} \int_0^\pi f(Re^{i\theta}) iRe^{i\theta} d\theta = 0 \quad (15)$$

$$\begin{aligned} -\lim_{\epsilon \rightarrow 0} \int_0^\pi f(\epsilon e^{i\theta}) i\epsilon e^{i\theta} d\theta &= -i \lim_{\epsilon \rightarrow 0} \int_0^\pi \frac{\epsilon e^{i\theta} \log(\epsilon e^{i\theta})}{\epsilon^2 e^{2i\theta} + a^2} d\theta \\ &= i \lim_{\epsilon \rightarrow 0} \int_0^\pi \frac{(e^{i\theta} \log(\epsilon) + ie^{i\theta} \theta)}{\epsilon e^{2i\theta} + \frac{a^2}{\epsilon}} d\theta \end{aligned}$$

As  $\epsilon \rightarrow 0$ ,  $a^2/\epsilon \rightarrow \infty$  faster than  $\log(\epsilon) \rightarrow -\infty$ , thus  $\frac{a^2}{\epsilon}$  dominates and the denominator goes to infinity faster than the numerator goes to  $-\infty$ , thus  $\frac{(e^{i\theta} \log(\epsilon) + ie^{i\theta} \theta)}{\epsilon e^{2i\theta} + \frac{a^2}{\epsilon}} \rightarrow 0$ . Thus,

Thus,

$$-\lim_{\epsilon \rightarrow 0} \int_0^\pi f(\epsilon e^{i\theta}) i\epsilon e^{i\theta} d\theta = 0 \quad (16)$$

Substitute 14, 15, and 16 into 8 and use the result from Cauchy's Formula.

Thus,

$$\begin{aligned} 2 \int_0^\infty f(x) dx + \frac{i\pi^2}{2a} &= \frac{\pi}{a} \left[ \log(a) + \frac{\pi}{2} i \right] \\ \int_0^\infty f(x) dx &= \frac{\pi \log(a)}{2a} \end{aligned}$$

□

### Problem 11a.

*Proof.* Let  $\mathbb{D} := \{z \in \mathbb{C} \mid |z| < 1\}$ . Slit the plane on  $(\infty, 0]$  and define the domain of the complex logarithm as  $\mathbb{C} - (-\infty, 0]$ .

Let  $\gamma$  be the contour along the unit circle.

Assume the contrary  $\exists z \in \mathbb{D} \ni 1 - az \leq 0$ , which means  $az \geq 1 \implies |az| \geq 1$ . Since  $|z| < 1$ , this means,  $|a| \geq 1$  which is a contradiction. Thus  $\forall z \in \mathbb{D} \ni 1 - az > 0 \implies \log(1 - az)$  has no poles on the unit disk and thus is holomorphic on the domain.

$z = 0$ , is a simple pole of  $\frac{\log(1-az)}{z}$  since  $z \frac{\log(1-az)}{z} = \log(1 - az)$  which is holomorphic on the domain.

By the Residue theorem,

$$\oint_\gamma \frac{\log(1-az)}{z} dz = 2\pi i \lim_{z \rightarrow 0} \frac{z \log(1-az)}{z} = 2\pi i \lim_{z \rightarrow 0} \log(1-az) = 0$$

Thus,

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = \frac{1}{i} \oint_\gamma \frac{\log(1-az)}{z} dz = 0$$

□

**Problem 11b.**

*Proof.* Suppose  $|a| = 1$  thus  $a = e^{i\varphi}$ .

Let  $\mathbb{D} := \{z \in \mathbb{C} \mid |z| < 1\}$ . Slit the plane on  $(\infty, 0]$  and define the domain of the complex logarithm as  $\mathbb{C} - (-\infty, 0]$ .

Let  $\gamma$  be the contour along the unit circle. We are allowed to integrate through  $z = 1$ , because  $z = 1$  is a logarithmic singularity.

Assume the contrary  $\exists z \in \mathbb{D} \ni 1 - z \leq 0$ , which means  $z \geq 1 \implies |z| \geq 1$  which is a contradiction. Thus  $\log(1 - z)$  is holomorphic on  $\mathbb{D}$ .

$z = 0$ , is a simple pole of  $\frac{\log(1-z)}{z}$  since  $z \frac{\log(1-z)}{z} = \log(1 - z)$  which is holomorphic on the domain.

By the Residue theorem,

$$\oint_{\gamma} \frac{\log(1-z)}{z} dz = 2\pi i \lim_{z \rightarrow 0} \frac{z \log(1-z)}{z} = 2\pi i \lim_{z \rightarrow 0} \log(1-z) = 0$$

Thus,

$$\begin{aligned} \int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta &= \int_0^{2\pi} \log |1 - e^{i\varphi} e^{i\theta}| d\theta \\ &= \int_0^{2\pi} \log |1 - e^{i(\theta+\varphi)}| d\theta \\ &= \int_{\varphi}^{2\pi+\varphi} \log |1 - e^{ix}| dx \quad (\text{By substitution } x = \theta + \varphi) \\ &= \int_{\varphi}^{2\pi} \log |1 - e^{ix}| dx + \int_{2\pi}^{2\pi+\varphi} \log |1 - e^{ix}| dx \\ &= \int_{\varphi}^{2\pi} \log |1 - e^{ix}| dx + \int_0^{\varphi} \log |1 - e^{ix}| dx \quad (\text{since } e^{ix} \text{ is periodic with period of } 2\pi) \\ &= \int_0^{2\pi} \log |1 - e^{ix}| dx \\ &= \frac{1}{z} \oint_{\gamma} \frac{\log(1-z)}{z} dz = 0 \end{aligned}$$

□

**Problem 12.**

*Proof.* Suppose  $u \notin \mathbb{Z}$ .

$\forall z \in \mathbb{C} \ni \sin(\pi z) = 0$ . Let  $z = a + bi$ . Thus,

$$\begin{aligned}
0 &= \sin(\pi z) \\
&= \frac{e^{iz} - e^{-iz}}{2i} \\
\implies 0 &= e^{i\pi z} - e^{-i\pi z} \\
&= e^{a\pi i} e^{-\pi b} - e^{-a\pi i} e^{\pi b} \\
\implies e^{a\pi i} e^{-\pi b} &= e^{-a\pi i} e^{\pi b} \implies -\pi b \pm a\pi i = \pi b \pm a\pi i \implies b = 0 \\
e^{a\pi i} &= e^{-a\pi i} \\
e^{2a\pi i} &= 1 \\
\implies \exists n \in \mathbb{Z} \ni 2\pi n = 2a\pi &\implies a \in \mathbb{Z}
\end{aligned}$$

Thus  $\forall n \in \mathbb{Z}$ ,  $n$  is zero of  $\sin(\pi z)$  and thus a pole of  $\cos(\pi z)/\sin(\pi z)$ .

To find out if  $n$  is a simple root of  $\cot$ , expand  $\cos(\pi z)$  and  $\sin(\pi z)$  around  $z = n$ .

$$\begin{aligned}
(z-n) \frac{\cos(\pi z)}{\sin(\pi z)} &= \frac{(z-n) \left[ (-1)^{n-1} + \frac{\pi^2}{2}(x-n)^2 - \frac{1}{4} \frac{\pi^4}{4!}(x-n)^4 + \dots \right]}{(z-n) - \frac{\pi^3}{3!}(z-n)^3 + \frac{\pi^5}{5!}(z-n)^5 + \dots} \\
&= \frac{(z-n) \left[ (-1)^{n-1} + \frac{\pi^2}{2}(x-n)^2 - \frac{1}{4} \frac{\pi^4}{4!}(x-n)^4 + \dots \right]}{(z-n) \left[ 1 - \frac{\pi^3}{3!}(z-n)^2 + \frac{\pi^5}{5!}(z-n)^4 + \dots \right]} \\
&= \frac{\left[ (-1)^{n-1} + \frac{\pi^2}{2}(x-n)^2 - \frac{1}{4} \frac{\pi^4}{4!}(x-n)^4 + \dots \right]}{\left[ 1 - \frac{\pi^3}{3!}(z-n)^2 + \frac{\pi^5}{5!}(z-n)^4 + \dots \right]}
\end{aligned}$$

$\frac{\left[ (-1)^{n-1} + \frac{\pi^2}{2}(x-n)^2 - \frac{1}{4} \frac{\pi^4}{4!}(x-n)^4 + \dots \right]}{\left[ 1 - \frac{\pi^3}{3!}(z-n)^2 + \frac{\pi^5}{5!}(z-n)^4 + \dots \right]}$  is holomorphic thus,  $n$  is a simple pole.

Let  $f(x) = \frac{\cot(\pi x)}{(x+u)^2}$ .

$z = -u$  is a order two pole of  $f$ . Thus  $f$  has poles at  $-u$  and every integer.

$\forall N \in \mathbb{N} \ni N > |u|$ , let  $\gamma_N$  be the contour which is a circle centered at the 0 with radius  $N + \frac{1}{2}$ .

$\forall n \in \mathbb{Z}$ ,

$$\begin{aligned}
\text{res}_n(f) &= \lim_{z \rightarrow n} (z-n) \frac{\cot(\pi z)}{(z+u)^2} \\
&= \lim_{z \rightarrow n} \frac{\cos(\pi z)(z-n)}{(z+u)^2 \sin(\pi z)} \\
&= \lim_{z \rightarrow n} \frac{\cos(\pi z) - (z-n) \sin(\pi z)}{2(z+u) \sin(\pi z) + (z+u)^2 \cos(\pi z)} \\
&= \frac{1}{z+u}
\end{aligned}$$

Similarly,

$$\begin{aligned}
\text{res}_{-u}(f) &= \lim_{z \rightarrow -u} \frac{d}{dz} (z+u)^2 \frac{\cot(\pi z)}{(z+u)^2} \\
&= \lim_{z \rightarrow u} \frac{d}{dz} \cot(\pi z) \\
&= -\pi \lim_{z \rightarrow u} \csc^2(\pi z) \\
&= -\pi \csc^2(\pi(-u)) = -\pi \csc^2(\pi u)
\end{aligned}$$

Thus by the residue formula,  $\forall N \in \mathbb{N} \ni N \geq |u|$

$$\begin{aligned}
\oint_{\gamma_N} f(z) dz &= 2\pi i \left[ \sum_{n=-N}^N \text{res}_n(f) + \text{res}_{-u}(f) \right] \\
&= 2\pi i \left[ \sum_{n=-N}^N \frac{1}{(n+u)^2} - \pi \csc^2(\pi u) \right]
\end{aligned}$$

$$\begin{aligned}
\lim_{N \rightarrow \infty} \oint_{\gamma} \frac{\pi \cot(\pi z)}{(z+u)^2} dz &= \lim_{N \rightarrow \infty} \pi \int_0^{2\pi} \frac{(N + \frac{1}{2})e^{i\theta} \cot(\pi(N + \frac{1}{2})e^{i\theta})}{((N + \frac{1}{2})e^{i\theta} + u)^2} dz \\
&=
\end{aligned}$$

Let  $M = N + \frac{1}{2}$

$$\begin{aligned}
\cot(\pi M e^{i\theta}) &= i \frac{e^{\pi i M e^{i\theta}} + e^{-\pi i M e^{i\theta}}}{e^{\pi i M e^{i\theta}} - e^{-\pi i M e^{i\theta}}} \\
&= i \frac{e^{\pi M i \cos(\theta)} e^{-\pi M \sin(\theta)} + e^{-\pi M i \cos(\theta)} e^{\pi M \sin(\theta)}}{e^{\pi M i \cos(\theta)} e^{-\pi M \sin(\theta)} - e^{-\pi M i \cos(\theta)} e^{\pi M \sin(\theta)}}
\end{aligned}$$

As  $N \rightarrow \infty$  sequentially (ie takes integer values),

If  $\theta \in (0, \pi)$

$$\cot(\pi M e^{i\theta}) \approx i \frac{e^{-\pi M i \cos(\theta)} e^{\pi M \sin(\theta)}}{-e^{-\pi M i \cos(\theta)} e^{\pi M \sin(\theta)}}$$

Thus  $|\cot(\pi M e^{i\theta})| \approx 1$  if  $\theta \in (\pi, 2\pi)$

If  $\theta \in (\pi, 2\pi)$ :

$$\cot(\pi M e^{i\theta}) \approx i \frac{e^{\pi M i \cos(\theta)} e^{-\pi M \sin(\theta)}}{e^{\pi M i \cos(\theta)} e^{-\pi M \sin(\theta)}}$$

Thus  $|\cot(\pi M e^{i\theta})| \approx 1$  if  $\theta \in (\pi, 2\pi)$

If  $\theta = n\pi$  for  $n \in [0, 1, 2]$ ,  $\cot((\pi M e^{i\theta})) = 0$ .

Thus  $|\cot(\pi M e^{i\theta})| \leq 1$

By the integral inequality,

$$\begin{aligned}
\left| \lim_{N \rightarrow \infty} \int_0^{2\pi} \frac{(N + \frac{1}{2})e^{i\theta} \cot(\pi(N + \frac{1}{2})e^{i\theta})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} dz \right| &\leq \lim_{N \rightarrow \infty} \int_0^{2\pi} \left| \frac{(N + \frac{1}{2})e^{i\theta} \cot(\pi(N + \frac{1}{2})e^{i\theta})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} \right| dz \\
&= \lim_{N \rightarrow \infty} \int_0^{2\pi} \left| \frac{(N + \frac{1}{2}) \cot(\pi(N + \frac{1}{2})e^{i\theta})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} \right| dz \\
&= \lim_{N \rightarrow \infty} \int_0^{2\pi} \left| \frac{(N + \frac{1}{2})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} \right| |\cot(\pi Me^{i\theta})| dz \\
&\leq \lim_{N \rightarrow \infty} \int_0^{2\pi} \left| \frac{(N + \frac{1}{2})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} \right| dz
\end{aligned}$$

As  $N \rightarrow \infty$ , the denominator dominates. thus,  $\lim_{N \rightarrow \infty} \int_0^{2\pi} \left| \frac{(N + \frac{1}{2})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} \right| dz = 0$

$$\left| \lim_{N \rightarrow \infty} \int_0^{2\pi} \frac{(N + \frac{1}{2})e^{i\theta} \cot(\pi(N + \frac{1}{2})e^{i\theta})}{((N + \frac{1}{2})e^{i\theta}) + u)^2} dz \right| = 0 \tag{17}$$

Thus,

$$2\pi i \left[ \lim_{N \rightarrow \infty} \sum_{n=-N}^N \frac{1}{(n+u)^2} - \pi \csc^2(\pi u) \right] = 0$$

Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+u)^2} = \pi \csc^2(\pi u) = \frac{\pi}{\sin(\pi u)}$$

□